

NutriAgent: Mid-Evaluation Group Discussion Guide

Structured Answers & Talking Points for the 15-Minute Panel Discussion (20 Marks)

Course: IT3041 Information Retrieval & Web Analytics | Institution: SLIIT | Format: 15-Min Group Discussion

Evaluation Goal: The panel evaluates conceptual clarity, system architecture, realistic planning, Responsible AI, and business value. Use this document to align all 4 group members with consistent, polished answers.

1. Why You Selected the Domain & Problem Statement

Real-World Problem:

- **Generic & Unsafe Online Advice:** Millions of people seek dietary advice online, but general search engines return generic or contradictory advice that ignores personal health restrictions.
- **Fatal Risks with Food Allergies:** Individuals with severe allergies (peanuts, shellfish, gluten) or conditions (diabetes, hypertension) risk severe harm if recommendations are not strictly filtered.
- **LLM Hallucination Problem:** Standard Large Language Models (e.g., ChatGPT) invent fake calorie counts and nutritional macros because they are trained to generate plausible text, not retrieve factual nutrition data.
- **High Cost of Professional Nutritionists:** Personalized consultation with clinical dietitians is unaffordable or inaccessible for many everyday individuals.

Who Experiences This Problem?

- Individuals with food allergies and chronic health conditions (celiac, diabetes, lactose intolerance).
- Fitness enthusiasts, athletes, and bodybuilders needing exact macronutrient targets (protein, carbs, fat).
- Busy professionals and families seeking quick, tailored, and healthy weekly meal plans.

Why is Agentic AI Suitable for Addressing It?

- A monolithic single AI model cannot handle security, NLP extraction, factual database retrieval, and constrained reasoning without failing or hallucinating.
- **Agentic Separation of Concerns:** By delegating specialized tasks to independent agents (e.g. deterministic hard-filtering in the IR Agent and explainable reasoning in the Planning Agent), we guarantee safety and factual accuracy.

2. Proposed System (What We Are Developing)

System Identity: NutriAgent — A Multi-Agent AI Nutrition Advisory Microservice System.

• **What it Does:** Accepts natural language input from a user (e.g., *'I want to build muscle, I'm allergic to peanuts, and I prefer vegetarian food'*), extracts dietary requirements, retrieves factual food records from official nutrition databases (USDA), and constructs an explainable, fact-checked meal plan.

• Core Value Proposition:

1. **Zero Hallucinated Numbers:** Every calorie and macro number comes from verified USDA food data.
2. **Guaranteed Safety:** Strict deterministic allergy filtering ensures allergen-containing items never reach the user.
3. **Explainability:** Every meal recommendation includes a transparent 'reason' citing user-specific goals and preferences.

3. Agents and Their Roles (Architecture & Communication)

Our system contains **4 independent microservices** connected in a synchronous HTTP chained architecture with a shared Pydantic JSON contract (shared/schemas.py):

Agent Name & Port	Why It Is Necessary	What It Does (Core Responsibilities)
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1. Security & Validation Agent (Port 8001)	Protects the system from malicious attacks and validates input before processing.	• System front door / API gateway. • JWT authentication & authorization. • Prompt injection defense & regex sanitization. • Rate limiting (slowapi) & trace-ID audit logging.
2. Intake & Profile Agent (Port 8002)	Converts unstructured human conversation into machine-readable user profiles.	• Natural Language Processing (NER) extraction. • Identifies user goals, allergies, medical conditions, and diet types. • Produces validated UserProfile JSON. • Coordinates downstream IR and Planning agents.
3. Nutrition IR Agent (Port 8003)	Acts as the 'Ground Truth' factual layer to prevent LLM hallucinations.	• Dense vector semantic search (Sentence-Transformers all-MiniLM-L6-v2 + ChromaDB). • Indexes USDA Foundation Foods (1,178 items). • Deterministic allergy & dietary hard-filtering. • Returns factual FoodItem objects with real macros.
4. Meal Planning Agent (Port 8004)	Synthesizes retrieved ingredients into practical, explainable meal plans.	• LLM reasoning layer (Claude / OpenAI SDK). • Prompt constrained strictly to retrieved items. • Generates transparent 'reason' strings per meal. • Outputs validated MealPlan structure.

How Agents Interact: Synchronous HTTP chain: Client ■■■ Security (8001) ■■■ Intake (8002) ■■■ calls IR (8003), then calls Planning (8004) ■■■ Response flows back: Planning ■■■ Intake ■■■ Security ■■■ Client.

4. Technical Implementation Plan

We have a clean, modular, and realistic technology stack:

- **Microservices Framework:** FastAPI (Python) running on independent ports (8001–8004) for lightweight, high-performance async HTTP communication.
- **Natural Language Processing (Intake):** spaCy Named Entity Recognition (NER) and regex entity extractors to extract allergies, goals, and conditions into Pydantic models.
- **Information Retrieval (IR):** Embedded chromadb vector database with sentence-transformers (all-MiniLM-L6-v2) for dense semantic retrieval over USDA datasets, paired with deterministic Python boolean filters for safety.
- **Generative AI / LLM Reasoning (Planning):** OpenAI / Anthropic Claude API with strict temperature control and prompt containment (forcing the model to only recommend items retrieved by the IR Agent).
- **Security & Cryptography:** pyjwt for token validation, cryptography.fernet for field-level health data encryption, and slowapi for DDoS/rate limiting.

5. Responsible AI (RAI) Plan

Addressing ethical and safety risks is central to NutriAgent's design:

Responsible AI Pillar	Potential Risk in Nutrition Domain	NutriAgent Mitigation Strategy
Explainability & Transparency	Black-box AI recommendations where users don't know why a food was chosen.	Every meal includes an explicit reason string tied to the user's profile and cites the factual data source (e.g. usda_fooddata_central).
Safety & Harm Prevention	Severe allergic reactions or dangerous advice for medical conditions.	Deterministic post-filtering guarantees allergens are excluded in code before reaching the LLM. Mandatory medical disclaimer: <i>'AI guidance, not medical advice.'</i>

Privacy & Data Protection	Exposure of sensitive health conditions, eating habits, and personal goals.	Symmetric encryption (Fernet) for stored health data, stateless request processing, and trace-ID logging without logging raw PII.
Fairness & Inclusivity	Biased food recommendations favoring only Western diets.	Dataset includes diverse international ingredients, vegetarian, vegan, and multicultural food options.

6. Commercialization & Business Plan

Target Users & Market Segments:

- **B2C Consumers:** Health-conscious individuals, fitness enthusiasts, bodybuilders, and people managing dietary restrictions.
- **B2B Professionals:** Dietitians, personal trainers, gym chains, and corporate wellness platforms needing automated meal plan creation.

Value Proposition:

- Reduces meal planning time from hours to seconds.
- Eliminates nutritional calculation errors with certified government nutrition databases.
- Provides enterprise-grade allergy safety and privacy compliance at a fraction of manual consultation costs.

Revenue & Pricing Models:

1. **B2C Freemium:** Free basic daily meal plan; **\$9.99/month Pro** for weekly grocery list generation, micronutrient tracking, and smart recipe substitutions.
2. **B2B SaaS Subscription:** **\$49–\$199/month** per clinic/gym for client management dashboard and branded white-label meal plan exports.
3. **B2B API Licensing:** Usage-based API pricing (\$0.02 per request) for third-party fitness and healthcare mobile apps.

Deployment & Market Introduction:

- Phase 1: Deploy microservices in lightweight Docker containers on cloud platforms (AWS / Render).
- Phase 2: Launch a mobile/web frontend (React / Flutter) with direct integration into wearable health APIs (Apple Health, Fitbit).

Key Takeaway for the 15-Min Evaluation: Remember that **NutriAgent** stands out because it solves the *LLM Hallucination & Safety problem* using an **Agentic architecture** with **Information Retrieval ground-truth data**. Emphasize how your microservices collaborate cleanly!